AS Business 9609 - Calculation Drills

Every quantitative skill in the AS syllabus, with repeated practice. Show all working and always state units ($, %, units). Worked answers are in the separate Calculation Drills - Answers booklet.

A. Added value (selling price - bought-in material cost)

A1. A bakery buys ingredients for $2 per loaf and sells each loaf for $5. Calculate the added value per loaf.

A2. A furniture maker buys $120 of materials per table and sells each table for $450. Calculate added value per table and total added value on 30 tables.

A3. A phone case costs $1.20 in materials and sells for $9.99. State the added value per case.

B. Labour turnover ((staff leaving / average staff) x 100)

B1. 24 staff left during the year; average staff employed was 160. Calculate labour turnover.

B2. A firm employs on average 480 people and 36 left last year. Calculate labour turnover.

B3. Labour turnover is 15% and average staff is 200. How many staff left?

C. Labour productivity (total output / number of employees)

C1. 18,000 units are produced by 45 workers. Calculate output per worker.

C2. A team of 12 assembles 3,600 units per week. Calculate weekly productivity per worker.

C3. Productivity is 250 units per worker and there are 60 workers. Calculate total output.

D. Market share ((firm sales / total market sales) x 100)

D1. A firm sells $6m in a market worth $48m. Calculate its market share.

D2. Total market sales are 500,000 units; the firm sells 90,000 units. Calculate market share.

D3. A firm has 12% share of a $250m market. Calculate its sales revenue.

E. Market growth ((change in market size / original size) x 100)

E1. A market grew from $80m to $92m. Calculate market growth.

E2. Market size fell from 400,000 to 360,000 units. Calculate the growth rate (state if negative).

E3. A $150m market grows by 8%. Calculate the new market size.

F. Capacity utilisation ((actual output / maximum output) x 100)

F1. Maximum output is 5,000 units; actual output is 3,750. Calculate capacity utilisation.

F2. A cinema has 250 seats; on average 175 are filled. Calculate capacity utilisation.

F3. A factory runs at 90% utilisation with maximum capacity of 20,000 units. Calculate actual output.

G. Working capital (current assets - current liabilities)

G1. Current assets $90,000; current liabilities $62,000. Calculate working capital.

G2. Current assets $45,000; current liabilities $51,000. Calculate working capital and comment.

H. Total cost & average cost

H1. Fixed costs $30,000; variable cost per unit $4; output 6,000 units. Calculate total variable cost, total cost and average cost per unit.

H2. Fixed costs $18,000; variable cost per unit $7; output 3,000 units. Calculate total cost and average cost.

I. Cash flow (net cash flow; closing = opening + net)

I1. Opening balance $8,000; inflows $30,000; outflows $34,000. Calculate net cash flow and closing balance.

I2. A firm has these months. Jan: open $2,000, inflow $18,000, outflow $15,000. Feb: inflow $16,000, outflow $20,000. Calculate the closing balance for Jan and for Feb.

J. Contribution, total contribution & profit

J1. Selling price $25; variable cost per unit $10. Calculate contribution per unit.

J2. Contribution per unit $15; units sold 4,000; fixed costs $40,000. Calculate total contribution and profit.

J3. Selling price $12, variable cost $7, fixed costs $25,000, units sold 7,000. Calculate profit.

K. Break-even, margin of safety & target profit

K1. Fixed costs $60,000; selling price $30; variable cost $18. Calculate contribution per unit and break-even output.

K2. Using K1, current output is 7,000 units. Calculate the margin of safety and the profit at that output.

K3. Fixed costs $45,000; selling price $20; variable cost $11. How many units must be sold to make a target profit of $18,000?

L. Variances (actual - budget; state favourable/adverse)

L1. Budgeted revenue $80,000; actual revenue $86,000. Calculate the variance and state its type.

L2. Budgeted costs $50,000; actual costs $57,000. Calculate the variance and state its type.

L3. Budgeted profit $30,000; actual profit $26,000. Calculate the profit variance and state its type.

END OF DRILLS - work them without a calculator first, then check.